

1. In the ideal gas equation, the gas constant R has the dimensions of :
(A) mole-atm K⁻¹ (B) litre mole
(C) litre-atm K⁻¹ mole⁻¹ (D) erg K⁻¹
2. Calculate the volume of 14 g nitrogen at S.T.P.
(A) 22.4 L (B) 5.6 L (C) 11.2 L (D) 44.8 L
3. For an ideal gas, number of moles per litre in terms of its pressure P, gas constant R and temperature T is :
(A) PT/R (B) PRT (C) P/RT (D) RT/P
4. How many moles of He gas occupy 22.4 litres at 30°C and one atmospheric pressure
(A) 0.90 (B) 1.11 (C) 0.11 (D) 1.0
5. At N.T.P. the volume of a gas is found to be 273 ml. What will be the volume of this gas at 600 mm Hg and 273°C.
(A) 391.8 mL (B) 380 ml (C) 691.6 ml (D) 750 ml
6. Which of the following is/are affected by electric field :
(A) Anode rays (B) Cathode rays (C) Both (A) and (B) (D) None of these
7. The ratio of e/m for a cathode ray :
(A) varies with a gas in a discharge tube (B) is fixed
(C) varies with different electrodes (D) is maximum if hydrogen gas is taken
8. Which of the following statements is wrong about anode rays :
(A) They travel in straight line (B) They produce heating effect
(C) They carry positive charge (D) None of these
9. Cathode ray are :
(A) Stream of electrons (B) Stream of α -particles
(C) Radiation (D) Stream of cations
10. Anode rays have :
(A) Mass only (B) Charge only
(C) No mass & no charge (D) Both Mass & Charge
11. Which is the correct statement about proton :
(A) It is a nucleus of deuterium (B) It is an ionised hydrogen molecule
(C) It is an ionised hydrogen atom (D) It is an α -particle

1. What is the ratio of mass of an electron to the mass of a proton
(A) 1 : 2 (B) 1 : 1 (C) 1 : 1837 (D) 1 : 3
2. Magnitude of e/m for O^- will be :
(A) 10^{-18} C/g (B) $10^{-20} N_A \text{ C/g}$ (C) $\frac{10^{12}}{N_A} \text{ C/g}$ (D) $\frac{10^{-18}}{N_A} \text{ C/g}$
3. Correct increasing order of e/m is :
(A) He^+, Ne^+, Ar^+ (B) Ar^+, Ne^+, He^+ (C) Ne^+, Ar^+, He^+ (D) All have same e/m
4. If two similar charges are brought closer, then PE of system will :
(A) first increase, then decrease (B) first decrease, then increase
(C) decreases continuously (D) increases continuously
5. Rutherford's experiment on scattering of particles showed for the first time that the atom has
(A) Electrons (B) Protons (C) Nucleus (D) Neutrons
6. Rutherford's alpha particle scattering experiment eventually led to the conclusion that
(A) Mass and energy are related
(B) Electrons occupy space around the nucleus
(C) Neutrons are buried deep in the nucleus
(D) The point of impact with matter can be precisely determined
7. When atoms are bombarded with alpha particles, only a few in million suffer deflection, others pass out undeflected. This is because
(A) The force of repulsion on the moving alpha particle is small
(B) The force of attraction on the alpha particle to the oppositely charged electrons is very small
(C) There is only one nucleus and large number of electrons
(D) The nucleus occupies much smaller volume compared to the volume of the atom

1. Pick up the correct statement about the following pair

I	II
Ar and S^{-2}	Fe and Ni^{2+}

(A) Both pairs contain isoelectronic species
 (B) Only pair (I) contains isoelectronic species
 (C) Only pair (II) contains isoelectronic species
 (D) None of the pair is isoelectronic

2. Which of the following atoms contains the least number of neutrons :

(A) ${}_{92}^{235}\text{U}$ (B) ${}_{92}^{238}\text{U}$ (C) ${}_{93}^{239}\text{Np}$ (D) ${}_{93}^{240}\text{Np}$

3. Which of the following species has more number of electrons in comparison with the neutrons.

(A) ${}_{13}^{27}\text{Al}^{3+}$ (B) ${}_1^2\text{H}$ (C) ${}_{17}^{37}\text{Cl}^{-}$ (D) ${}_{15}^{31}\text{P}^{-3}$

4. Identify isoelectronic pair(s) :

(A) CO_3^{2-} and NO_3^{-} (B) PO_4^{3-} and SO_4^{2-}
 (C) CO and N_2 (D) All of these

5. A^{+5} is isoelectronic with N^{3-} . If number of neutron in A is equal to $Z + 1$, then GAM of A will be :

(A) 30 g (B) 29 g (C) 31 g (D) 32 g

6. An element is represented by ${}_Z\text{X}^A$. If $A = 2Z + 1$ and X^{+} is isoelectronic with Si^{+4} , then which is(are) correct:

(A) X is Na atom (B) GAM of X is 23 g. (C) X has 12 neutrons. (D) All are correct.

7. Which of the following is a pair of isotones

(A) Ar^{38} and K^{40} (B) Cl^{37} and K^{39} (C) K^{40} and Ca^{40} (D) K^{39} and K^{40}

8. If velocity of α -particle is made $1/3$ times, then % variation in distance of closest approach will be : (with respect to initial)

(A) 300% (B) 600% (C) 800% (D) 80%

9. Distance of closest approach of α -particle when projected to Ag ($Z = 47$) nucleus will be :

(A) $\frac{47e^2}{m_{\alpha} v_{\alpha}^2}$ (B) $\frac{108e^2}{m_{\alpha} v_{\alpha}^2}$ (C) $\frac{108e^2}{\pi \epsilon_0 m_{\alpha} v_{\alpha}^2}$ (D) $\frac{47e^2}{\pi \epsilon_0 m_{\alpha} v_{\alpha}^2}$

* Marked Questions are having one or more than one correct option.

1. How long would it take a radio wave of frequency $6 \times 10^3 \text{ sec}^{-1}$ to travel from mars to the earth, a distance of $8 \times 10^7 \text{ km}$: (speed of EMR = $3 \times 10^8 \text{ m/s}$)
(A) 2 min 46 sec. (B) 4 min 26 sec. (C) 10 min 46 sec. (D) none of these
2. Wave number of radiations having frequency of $4 \times 10^4 \text{ Hz}$ will be :
(A) $1.33 \times 10^{-6} \text{ cm}^{-1}$ (B) $1.33 \times 10^{-7} \text{ cm}^{-1}$ (C) $9 \times 10^{-11} \text{ cm}^{-1}$ (D) $4 \times 10^{-5} \text{ cm}^{-1}$
3. Correct order(s) is(are) :
(A) X-rays < UV < Radio(energy) (B) Radio < Micro < Infra red (wave number)
(C) Cosmic < γ -rays < X-rays (frequency) (D) All are correct
- 4*. Which is correct order of wavelength (increasing or decreasing) :
(A) Cosmic, Gamma, X-rays (B) X-rays, UV, Visible
(C) Red, Orange, Blue (D) Infra red, Micro, Radio
5. The value of Planck's constant is $6.63 \times 10^{-34} \text{ Js}$. The velocity of light is $3.0 \times 10^8 \text{ ms}^{-1}$. Which value is closest to the wavelength in nanometers of a quantum of light with frequency of $8 \times 10^{15} \text{ s}^{-1}$?
(A) 2×10^{-25} (B) 5×10^{-18} (C) 4×10^1 (D) 3×10^7
6. A 100 watt bulb emits electromagnetic radiations of wavelength 400 nm. Then number of photons emitted per second by the bulb will be :
(A) 2×10^{20} (B) 2×10^{15} (C) 2×10^{10} (D) 2×10^5
7. Number of photons of 4000 pm that provide 1 Joule energy, will be :
(A) 10^{16} (B) 5×10^{16} (C) 2×10^{16} (D) 9×10^{16}
8. A dye absorbs n_1 photons of wavelength $4000 \text{ \AA}$ and emits n_2 photons of wavelength $5000 \text{ \AA}$. If 40% of absorbed energy is emitted, then $\frac{n_1}{n_2}$ will be :
(A) 1 (B) 2 (C) 3 (D) 4

1. One quantum is absorbed per gaseous molecule of X_2 for converting it into X atoms. If light absorbed has wave length $1240\text{\AA}$, then bond energy of X_2 will be :
- (A) 10 eV/molecule (B) 20 J/mole (C) 48 eV/molecule (D) 184 J/mol
2. Momentum of a photon of wavelength λ is :
- (A) $\frac{h}{\lambda}$ (B) zero (C) $\frac{h\lambda}{c^2}$ (D) $\frac{h\lambda}{c}$
3. As the momentum associated with an electron decreases, its de-Broglie wavelength :
- (A) Decreases (B) Increases
(C) Remains unchanged (D) Cannot be predicted
4. The de-Broglie wavelength of a particle with mass 1 g and velocity 100 m/s is :
- (A) 6.63×10^{-33} m (B) 6.63×10^{-34} m (C) 6.63×10^{-35} m (D) 6.65×10^{-36} m
5. If the Planck's constant, $h = 6.6 \times 10^{-34}$ Js, the de-Broglie wavelength of a particle having momentum 3.3×10^{-24} Kg ms⁻¹ will be
- (A) $0.02\text{\AA}$ (B) $90.5\text{\AA}$ (C) $2\text{\AA}$ (D) $500\text{\AA}$
6. Wavelength associated with an electron moving with a velocity of 10^3 msec⁻¹ will be :
- (A) 7.25×10^{-9} m (B) 7.25×10^{-11} m (C) 7.25×10^{-7} m (D) none of these
7. Calculate ratio of de-Broglie wavelength for a proton and α -particle, if their kinetic energies are same :
- (A) 3 : 1 (B) 2 : 1 (C) 1 : 3 (D) 1 : 2
8. For an electron $m \times KE = h$ (using SI units). Then wavelength associated with electron will be (m is mass of electron) :
- (A) $\frac{h}{3}$ (B) $\sqrt{\frac{h}{2}}$ (C) $\sqrt{\frac{h}{3}}$ (D) $\sqrt{\frac{2h}{3}}$



* Marked Questions are having one or more than one correct option.

1. A 0.66 kg ball is moving with a speed of 100 m/s. The associated wavelength will be : ($h = 6.6 \times 10^{-34}$ Js)
(A) 6.6×10^{-32} m (B) 6.6×10^{-34} m (C) 1.0×10^{-35} m (D) 1.0×10^{-32} m
2. The de-broglie wavelength associated with a ball of mass 1 kg having kinetic energy 0.5 J is.
(A) 6.626×10^{-34} m (B) 13.20×10^{-34} m (C) 10.38×10^{-21} m (D) 6.626×10^{-34} Å

Comprehension (Q. 3 to 7) :

Metal	Li	Na	K	Mg	Cu
(work function in eV)	2.4	2.3	2.2	3.7	4.8

3. How many metals can show photoelectric effect by using photons of wavelength 4000 Å.
(A) 4 (B) 5 (C) 2 (D) 3
- 4*. Cu metal can show photoelectric effect if approximate wavelength of photon is(are) :
(A) $\lambda = 5000$ Å (B) $\lambda = 2500$ Å (C) $\lambda = 2000$ Å (D) $\lambda = 4000$ Å
5. If photon of wavelength 4000 Å is used, then highest value of stopping potential can be for :
(A) Li (B) Na (C) K (D) All will have same value
6. $KE_{\max}$ of photoelectron using photon of wavelength 2000 Å at Cu metal will be :
(A) 2.4 eV (B) 1.4 eV (C) 1.9 eV (D) 3.4 eV
- 7*. On increasing intensity of EMR of wavelength 3000 Å, which metal will show increment in photo current :
(A) Na (B) Mg (C) Cu (D) Li
8. When the frequency of light incident on a metallic plate is doubled, the maximum KE of the emitted photoelectrons will be :
(A) Doubled (B) Halved
(C) Increased but more than double of the initial maximum KE (D) Unchanged

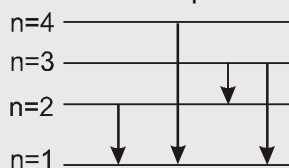
- The energy absorbed by each molecule (A_2) of a substance is 4.4×10^{-19} J and both energy per molecule is 4.0×10^{-19} J. The kinetic energy of the molecule per atom will be :
 (A) 2.0×10^{-20} J (B) 2.2×10^{-19} J (C) 2.0×10^{-19} J (D) 4.0×10^{-20} J
- The difference in angular momentum associated with the electron in two successive orbits of hydrogen atom is :
 (A) h/π (B) $h/2\pi$ (C) $h/2$ (D) $(n-1)h/2\pi$
- The expression for Bohr's radius of an atom is
 (A) $r = \frac{n^2 h^2}{4\pi^2 k m e^4 Z^2}$ (B) $r = \frac{n^2 h^2}{4\pi^2 k m z e^2}$ (C) $r = \frac{n^2 h^2}{4\pi^2 k m e^2 Z^2}$ (D) $r = \frac{n^2 h^2}{4\pi^2 k m^2 e^2 Z^2}$
- If r_1 is the radius of the first orbit of hydrogen atom, then the radii of second, third and fourth orbits in terms of r_1 are :
 (A) r_1^2, r_1^3, r_1^4 (B) $8r_1, 27r_1, 64r_1$ (C) $4r_1, 9r_1, 16r_1$ (D) $2r_1, 6r_1, 8r_1$
- The ratio of radii of second orbits of He^+ , Li^{2+} and Be^{3+} is :
 (A) 1 : 2 : 3 (B) 6 : 4 : 3 (C) 3 : 4 : 6 (D) none of these
- The radius of hydrogen atom is $0.53 \text{ \AA}$. The radius of ${}_3Li^{+2}$ is of :
 (A) $1.27 \text{ \AA}$ (B) $0.17 \text{ \AA}$ (C) $0.57 \text{ \AA}$ (D) $0.99 \text{ \AA}$
- The electronic velocity in the fourth Bohr's orbit of hydrogen atom is u . The velocity of the electron in the first orbit of same atom would be :
 (A) $u/4$ (B) $16u$ (C) $u/16$ (D) $4u$
- If the speed of electron in second orbit of He^+ is " v ", then the speed of electron in first Bohr orbit of hydrogen atom will be :
 (A) $x/2$ (B) $2x$ (C) x (D) $4x$



- Of the following, which statement(s) regarding Bohr theory is/are wrong :
 (A) Kinetic energy of an electron is half of the magnitude of its potential energy.
 (B) Kinetic energy of an electron is negative of total energy of electron.
 (C) Energy of electron decreases with increase in the value of the orbit number.
 (D) The ionization energy of H-atom in the first excited state is negative of one fourth of the energy of an electron in the ground state.
- The kinetic energy of an electron in a H-like species is :
 (A) Directly proportional to n^2 (where n = orbit number)
 (B) Inversely proportional to Z^2 (where Z = Atomic number)
 (C) Both (A) & (B)
 (D) None of these
- In an atom, two electrons move around the nucleus in circular orbits of radii R and $4R$. The ratio of the time taken by them to complete one revolution is : (Assume Bohr's model to be valid)
 (A) 1 : 4 (B) 4 : 1 (C) 1 : 8 (D) 8 : 7
- The energy level for ${}_ZA^{(+Z-1)}$ can be given by :
 (A) E_n for $A^{(+Z-1)} = Z^2 \times E_n$ for H (B) E_n for $A^{(+Z-1)} = Z \times E_n$ for H
 (C) E_n for $A^{(+Z-1)} = \frac{1}{Z^2} \times E_n$ for H (D) E_n for $A^{(+Z-1)} = \frac{1}{Z} \times E_n$ for H
- In hydrogen atom, energy of second excited state is -1.51 eV. Then, KE of same orbit of hydrogen atom is:
 (A) + 3.4 eV (B) + 1.51 eV (C) -13.6 eV (D) + 13.6 eV
- The energy of the fourth Bohr orbit of the hydrogen atom is -82 kJ mol^{-1} ; hence the energy of second Bohr orbit would be
 (A) $-1312 \text{ kJ mol}^{-1}$ (B) -328 kJ mol^{-1} (C) -41 kJ mol^{-1} (D) -164 kJ mol^{-1}
- The ratio of the difference in energy of electron between the first and second Bohr's orbit to that between second and third Bohr's orbit is :
 (A) $1/3$ (B) $27/5$ (C) $9/4$ (D) $4/9$

* Marked Questions are having one or more than one correct option.

- If the binding energy of 3rd orbit of a H-like species is 13.6 eV, then the species must be :
(A) Be³⁺ (B) Li⁺ (C) He⁺ (D) None of these
- If the wavelength of photon emitted from an electron jump $n = 4$ to $n = 2$ in a H-like species is 1216 Å, then the species is :
(A) H-atom (B) He⁺ ion (C) Li²⁺ ion (D) Be³⁺ ion
- If the I excitation potential of a hypothetical H-like atom is 10.2x V, then the value of II excitation energy is about :
(A) 13.6x eV (B) 12.09x eV (C) -10.2x eV (D) 40.5x eV
- The highest excited state that an unexcited hydrogen atom sample can reach when it is bombarded with photons of 12.2 eV, will be :
(A) $n = 4$ (B) $n = 2$ (C) $n = 3$ (D) No transition will occur.
- Considering Bohr model to be valid, the time period of revolution of electron revolving in n^{th} orbit in aH-like species (atomic number = Z) is proportional to $\frac{n^p}{Z^q}$. Find the value of (p + q) :
(A) 4 (B) -5 (C) -4 (D) 5
- If the binding energy of first excited state of a hydrogen like sample is 54.4 eV, then determine the atomic number Z of the H-like species :
(A) 1 (B) 2 (C) 3 (D) 4
- Suppose that a hypothetical H-like atom gives a red, green, blue and violet line in spectrum. Which jump, according to figure, would give off the violet spectral line :



- (A) $3 \rightarrow 1$ (B) $2 \rightarrow 1$ (C) $4 \rightarrow 1$ (D) $3 \rightarrow 2$

- 8.* Which of the following is/are correct according to Bohr model for H-like species (atomic number = Z):

- (A) Velocity of electron in n^{th} orbit $\propto \frac{Z}{n}$
 (B) Frequency of revolution of electron in n^{th} orbit $\propto \frac{Z^2}{n^3}$
 (C) de-Broglie wavelength of electron revolving in n^{th} orbit $\propto \frac{n}{Z}$
 (D) | Energy | of electron in n^{th} orbit $\propto \frac{n^2}{Z^2}$

1. Paschen series for H-atom lies in _____ region of electromagnetic spectrum :
(A) Visible (B) Ultra-violet (C) Infra red (D) Cannot be predicted
2. The longest λ for the Lyman series for H is : (Given $R_H = 109678 \text{ cm}^{-1}$)
(A) 1215 Å (B) 1315 Å (C) 1415 Å (D) 1515 Å
3. The shortest for the Brakett series for H is : (Given $R_H = 109678 \text{ cm}^{-1}$)
(A) 1459 Å (B) 4052 Å (C) 4052 nm (D) 1459 nm
4. The wavelength of a spectral line for an electronic transition is inversely related to :
(A) No. of electrons undergoing transition
(B) The nuclear charge of the atom
(C) The velocity of an electron undergoing transition
(D) The difference in the energy levels involved in the transition
5. Line spectrum is characteristic of :
(A) Molecule (B) Atom (C) Both (D) None of these
6. Different lines in Lyman series of hydrogen spectrum lie in :
(A) UV (B) Visible (C) IR (D) None of these
7. The spectral lines corresponding to the radiation emitted by an electron jumping from 6th, 5th and 4th orbits to second orbit belong to :
(A) Lyman series (B) Balmer series
(C) Paschen series (D) Pfund series
8. The speed of the electron in the 1st orbit of the hydrogen atom in the ground state is (c is the velocity of light):
(A) $\frac{c}{1.37}$ (B) $\frac{c}{1370}$ (C) $\frac{c}{13.7}$ (D) $\frac{c}{137}$
9. Which one of following explains light both as a stream of particles and as wave motion ?
(A) Diffraction (B) $\lambda = \frac{h}{p}$ (C) Interference (D) Photoelectric effect

1. The number of spectral lines that can be possible when electrons in 7th shell in different hydrogen atoms return to the 2nd shell.
(A) 12 (B) 15 (C) 14 (D) 10
2. If ionisation energy of hydrogen be E, then the ionisation energy of He^+ would be :
(A) E (B) 2E (C) 4E (D) 0.5 E
3. The energy of second orbit of hydrogen is equal to the energy of :
(A) Fourth orbit of He^+ (B) Fourth orbit of Li^{2+}
(C) Second orbit of He^+ (D) Second orbit of Li^{2+}
4. Which of the following electron transitions in hydrogen atom will require largest amount of energy ?
(A) From $n = 1$ to $n = 2$ (B) From $n = 2$ to $n = 1$
(C) From $n = \infty$ to $n = 1$ (D) From $n = 3$ to $n = 5$
5. Wave number of a spectral line for a given transition is $x \text{ cm}^{-1}$ for Be^{+3} , then its value for He^+ for same transition is :
(A) $x \text{ cm}^{-1}$ (B) $4x \text{ cm}^{-1}$ (C) $\frac{x}{4} \text{ cm}^{-1}$ (D) $2x \text{ cm}^{-1}$
6. Change in angular momentum when an electron makes a transition corresponding to the 3rd line of the Balmer series in Li^{2+} ion is :
(A) $\frac{h}{2\pi}$ (B) $\frac{2h}{2\pi}$ (C) $\frac{3h}{2\pi}$ (D) $\frac{4h}{2\pi}$
7. If the series limit of wavelength of the Lyman series for the hydrogen atom is $912 \text{ \AA}$, then the series limit of wavelength for the Balmer series of the hydrogen atom is :
(A) $912 \text{ \AA}$ (B) $912 \times 2 \text{ \AA}$ (C) $912 \times 4 \text{ \AA}$ (D) $912/2 \text{ \AA}$
8. If the shortest wavelength of H-atom in Lyman series is x, then longest wavelength in Balmer series of He^+ is :
(A) $\frac{9x}{5}$ (B) $\frac{36x}{5}$ (C) $\frac{x}{4}$ (D) $\frac{5x}{9}$

1. Ionisation energy of hydrogen atom is 13.6 eV. Hydrogen atom in the ground state are excited by monochromatic light of energy 12.1 eV. The maximum number of spectral lines emitted by hydrogen according to Bohr's theory will be:
(A) One (B) Two (C) Three (D) Four
2. What transition in He^+ ion shall have the same wave number as the first line in Balmer series of hydrogen atom :
(A) $3 \longrightarrow 2$ (B) $6 \longrightarrow 4$ (C) $5 \longrightarrow 3$ (D) $7 \longrightarrow 5$
3. Heisenberg's uncertainty (which is $\Delta x \cdot \Delta p = \frac{h}{4\pi}$) can be further reduced by :
(A) using advanced instruments (B) using photon of higher energy
(C) using photon of lower wave number (D) it can not be reduced by any mean
4. If uncertainty in position and velocity of a particle are 10^{-10} m and $5.27 \times 10^{-24} \text{ msec}^{-1}$, then mass of particle will be :
(A) 0.1 g (B) 0.1 Kg (C) 10^3 Kg (D) 10^{-3} g
5. A golf ball has a mass of 40 g and a speed of 45 m/s. If the speed can be measured within accuracy of 2%, then uncertainty in position will be approximately :
(A) 10^{-33} m (B) 10^{-40} m (C) 10^{-20} m (D) 10^{-50} m
6. The uncertainty in momentum of an electron is $1 \times 10^{-5} \text{ kg.m/s}$. The uncertainty in its position will be ($h = 6.62 \times 10^{-34} \text{ kg m}^2/\text{s}$) :
(A) 1.05×10^{-28} m (B) 1.05×10^{-26} m
(C) 5.27×10^{-30} m (D) 5.25×10^{-28} m
7. The measurement of the electron position is associated with an uncertainty in momentum which is equal to $1 \times 10^{-18} \text{ g cm s}^{-1}$. The Uncertainty in electron velocity is, (mass of an electron is $9 \times 10^{-28} \text{ g}$)
(A) $1.11 \times 10^9 \text{ cm s}^{-1}$ (B) $1.11 \times 10^6 \text{ cm s}^{-1}$
(C) $1.11 \times 10^5 \text{ cm s}^{-1}$ (D) $1.11 \times 10^{11} \text{ cm s}^{-1}$

* Marked Questions are having one or more than one correct option.

- The following quantum numbers are possible for how many orbital (s) $n = 4, l = 2, m = +1$:
(A) 1 (B) 2 (C) 3 (D) 4
- Which of the following sets of quantum numbers is not possible:

	n	l	m	s		n	l	m	s
(A)	2	2	1	+1/2	(B)	4	3	-2	-1/2
(C)	1	0	-1	-1/2	(D)	3	2	3	+1/2
- What is the maximum numbers of electrons that can be associated with the following set of quantum numbers?
 $n = 4, l = 2$ and $m = -2$.
(A) 6 (B) 4 (C) 2 (D) 10
- Any d-orbital can accommodate up to :
(A) 2 electrons (B) 6 electrons (C) 10 electrons (D) 14 electrons
- Two electrons in K-shell will differ in :
(A) principal quantum number (B) spin quantum number
(C) azimuthal quantum number (D) magnetic quantum number
- Total number of "m" values for $n=3$
(A) 3 (B) 6 (C) 9 (D) 18
- An electron has a spin quantum number $s = +1/2$ and a magnetic quantum number -1 . It cannot be present in :
(A) d-orbital (B) f-orbital (C) s-orbital (D) p-orbital
- For a shell of principal quantum number $n = 4$, there are :
(A) 16 orbitals (B) 4 subshells
(C) maximum 32 electrons (D) 4 electrons with $l = 3$

* Marked Questions are having one or more than one correct option.

- Which orbital has two angular nodal planes.
(A) s (B) p (C) d (D) f
- The maximum number of 3d-electrons that can have $s = -\frac{1}{2}$, are
(A) 10 (B) 3 (C) 5 (D) 7
- The maximum number of electrons possible in a subshell is equal to
(A) $2\ell + 1$ (B) $2n^2$ (C) ℓ^2 (D) $4\ell + 2$
- Which set has the same number of s-electrons
(A) C, Cu^{2+} , Zn (B) Cu^{2+} , Fe^{2+} , Ni^{2+} (C) S^{2-} , Ni^{2+} , Zn (D) None of these
- Which d-orbital does not have four lobes
(A) $d_{x^2-y^2}$ (B) d_{xy} (C) d_{z^2} (D) d_{xz}
- Which electronic configuration does not follow the Pauli's exclusion principle
(A) $1s^2, 2s^2 2p^4$ (B) $1s^2, 2s^2 2p^4 3s^2$ (C) $1s^2, 2p^4$ (D) $1s^2, 2s^2 2p^6 3s^3$
- For a d-electron, the orbital angular momentum is
(A) $\sqrt{6}\hbar$ (B) $\sqrt{2}\hbar$ (C) $\hbar$ (D) $2\hbar$
- * In hydrogen atom, which energy level order is/are correct.
(A) $1s < 2p$ (B) $2p = 2s$ (C) $2p > 2s$ (D) $2p < 3s$

* Marked Questions are having one or more than one correct option.

- The number of neutrons emitted when $^{235}_{92}\text{U}$ undergoes controlled number fission to $^{142}_{54}\text{Xe}$ and $^{90}_{38}\text{Xe}$
(A) 2 (B) 3 (C) 5 (D) 0
- In the transformation of $^{238}_{92}\text{U}$ to $^{234}_{92}\text{U}$, if one emission is an α -particle, what should be the other emission's
(A) Two β^- (B) Two β^- and one β^+
(C) One β^- and one γ (D) One β^+ and one β^-
- The sequence of the following nuclear reaction

$$^{238}_{98}\text{U} \xrightarrow{-\alpha} \text{y} \xrightarrow{-\beta} \text{z} \xrightarrow{-\beta} \text{L} \xrightarrow{n\alpha} ^{218}_{90}\text{U}$$
 What is the value of "n".
(A) 3 (B) 4 (C) 5 (D) 6
- Which of the following notations show the product correctly.
(A) $^{242}_{96}\text{Cm}(\alpha, 2n) ^{243}_{97}\text{Bk}$ (B) $^{10}_5\text{B}(\alpha, n) ^{13}_7\text{N}$
(C) $^{14}_7\text{N}(n, p) ^{14}_6\text{N}$ (D) $^{28}_{14}\text{Si}(d, n) ^{29}_{15}\text{P}$
- $^{238}_{92}\text{U}$ emits 8 α -particles and 6 β -particles. The proton/neutron ratio in the product nucleus is
(A) $\frac{41}{60}$ (B) $\frac{40}{61}$ (C) $\frac{41}{62}$ (D) $\frac{42}{61}$
- Consider the following nuclear reactions

$$^{238}_{92}\text{M} \longrightarrow ^x_y\text{N} + 2^4_2\text{He}$$

$$^x_y\text{N} \longrightarrow ^A_B\text{L} + 2\beta^+$$
 The number of neutrons in element L is
(A) 142 (B) 144 (C) 140 (D) 146
- Among the following nuclides, the highest tendency to decay by β^+ emission.
(A) ^{59}Cu (B) ^{63}Cu (C) ^{67}Cu (D) ^{68}Cu
- Which electronic configuration is not possible ?
(A) $2p^6$ (B) $2d^5$ (C) $3s^1$ (D) $4f^{12}$
- Which has highest number of unpaired electrons ?
(A) Mn (B) Mn^{5+} (C) Mn^{3+} (D) Mn^{4+}
- * The probability of finding an electron in a P_x orbital is zero
(A) In the yz-plane (B) In the xy-plane
(C) In the y-direction (D) In the z-direction

ANSWER KEY

DDT – 05

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|---------|--------|--------|--------|---------|
| 1. (C) | 2. (C) | 3. (C) | 4. (A) | 5. (C) |
| 6. (C) | 7. (B) | 8. (D) | 9. (A) | 10. (D) |
| 11. (C) | | | | |

DDT – 06

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|--------|--------|--------|--------|--------|
| 1. (C) | 2. (B) | 3. (B) | 4. (C) | 5. (C) |
| 6. (B) | 7. (D) | | | |

DDT – 07

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|--------|--------|--------|--------|--------|
| 1. (A) | 2. (A) | 3. (D) | 4. (D) | 5. (C) |
| 6. (D) | 7. (B) | 8. (C) | 9. (D) | |

DDT – 08

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|--------|--------|--------|------------|--------|
| 1. (B) | 2. (A) | 3. (B) | 4*. (ABCD) | 5. (C) |
| 6. (A) | 7. (C) | 8. (B) | | |

DDT – 09

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|--------|--------|--------|--------|--------|
| 1. (A) | 2. (A) | 3. (B) | 4. (A) | 5. (A) |
| 6. (C) | 7. (B) | 8. (B) | | |

DDT – 10

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|--------|-----------|--------|----------|--------|
| 1. (C) | 2. (A) | 3. (D) | 4*. (BC) | 5. (C) |
| 6. (B) | 7*. (ABD) | 8. (C) | | |

DDT – 11

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|--------|--------|--------|--------|--------|
| 1. (A) | 2. (B) | 3. (B) | 4. (C) | 5. (B) |
| 6. (B) | 7. (D) | 8. (C) | | |

DDT – 12

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|--------|--------|--------|--------|--------|
| 1. (C) | 2. (D) | 3. (C) | 4. (A) | 5. (B) |
| 6. (B) | 7. (B) | | | |

DDT – 13

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|--------|--------|-----------|--------|--------|
| 1. (B) | 2. (B) | 3. (B) | 4. (C) | 5. (D) |
| 6. (D) | 7. (C) | 8.* (ABC) | | |

DDT – 14

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|--------|--------|--------|--------|--------|
| 1. (C) | 2. (A) | 3. (D) | 4. (D) | 5. (B) |
| 6. (A) | 7. (B) | 8. (D) | 9. (B) | |

DDT – 15

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|--------|--------|--------|--------|--------|
| 1. (B) | 2. (C) | 3. (A) | 4. (A) | 5. (C) |
| 6. (C) | 7. (C) | 8. (A) | | |

DDT – 16

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|--------|--------|--------|--------|--------|
| 1. (C) | 2. (B) | 3. (D) | 4. (B) | 5. (A) |
| 6. (C) | 7. (A) | | | |

DDT – 17

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|--------|-----------|----------|--------|--------|
| 1. (A) | 2.* (ACD) | 3. (C) | 4. (A) | 5. (B) |
| 6. (C) | 7. (C) | 8. (ABC) | | |

DDT – 18

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|--------|--------|----------|--------|--------|
| 1. (C) | 2. (C) | 3. (D) | 4. (B) | 5. (C) |
| 6. (D) | 7. (A) | 8.* (AD) | | |

DDT – 19

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|--------|--------|--------|--------|------------|
| 1. (B) | 2. (A) | 3. (B) | 4. (B) | 5. (C) |
| 6. (B) | 7. (A) | 8. (B) | 9. (A) | 10.* (ACD) |